

Gordon Cheuk

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TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript

Web & Frameworks: HTML, CSS, Node.js, React

Libraries & Tools: Git, GitHub, NumPy, REST APIs, JSON, PyTorch

Concepts: Data Structures, Algorithm Design, Memory Management, Linear Algebra, Systems Programming, Probability & Simulation, Fundamentals of AI

PROJECTS

Reversi AI — Python | PyTorch, NumPy, Pygame

Sept 2025 - Dec 2025

- Implemented a Minimax game-playing agent with alpha-beta pruning, improving decision-time efficiency by **60–70%** compared to the unpruned version.
- Designed a weighted heuristic (disc differential, mobility, corner control) that boosted performance by **~185% at depth 2**, demonstrating the impact of corner priorities
- Built automated benchmarking scripts to run **100+ simulated games**, collecting win rates, score differentials, and victory margins against Random, Greedy, and multi-depth Minimax agents.
- Validated depth scaling: deeper search improved outcomes by **25–35% per depth increase**, with higher-depth agents winning **100%** of head-to-head depth-vs-depth matchups.

Custom Memory Allocator — C

- Developed a **custom memory allocator** in C from scratch, implementing **malloc()**, **free()**, and **calloc()** to manage a static memory pool.
- Benchmarked allocator performance with **First-Fit**, **Best-Fit**, and **Worst-Fit strategies**, implementing **memory coalescing** to optimise block reuse across all approaches.
- Gained deep understanding of **heap management**, **fragmentation trade-offs**, and **low-level memory operations**

ClearWrite — JavaScript | OpenAI, TextGears, ProWritingAid APIs

- Developed a full-stack React/Node.js web app integrating **TextGears** and **Sapling APIs** for grammar checking, AI detection, rephrasing, readability, summarization, and sentiment analysis.
- Designed and implemented the **frontend interface**, including dynamic sections, buttons, hover effects, animations, and responsive layouts to enhance user interaction and display of results.
- Managed **API integration** for the Evaluation Page, ensuring reliable retrieval, processing, and presentation of data from multiple external APIs.
- Created **custom styles and UI enhancements** for tone analysis, sentiment scores, and original text display, improving clarity and usability.

Study Buddy (Hackathon)— Web App | ElevenLabs, Google Gemini, Node.js, React

- Built a **full-stack AI tutoring platform**, integrating text-to-speech and AI-generated responses for interactive study sessions.
- Designed and implemented **multiple tutor personas** via **prompt engineering**, tailoring explanations to diverse audiences, including students, gamers, and casual learners.
- Crafted **dynamic prompts** to personalize explanations, increasing engagement and comprehension across different learning styles.

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WORK EXPERIENCE

Wu's Taekwondo School | Richmond, BC

Aug. 2019 - July. 2025

National Level Taekwondo Coach

- Coach athletes in advanced techniques, acrobatics, and strategy, fostering skill development and **confidence under pressure**.
- Design and lead practice sessions with **individualised training plans** tailored to athlete goals and performance levels.
- Provide **emotional support** and **motivation**, helping athletes manage **stress and competition nerves**.
- Track progress with **detailed performance notes**, identify **areas for improvement**, and guide **growth**.
- Recognised as a **national-level coach**, mentoring athletes for regional and international competitions.

Flight Club Artistic Movement | Richmond, BC

Dec. 2022 - Present

Instructor

- Taught adults and children in **tricking**, a combination of **martial arts and acrobatics**, including students in **film, dance, stunts, and martial arts**.
- Planned and led workshops, managing logistics and creating **customised training programs** for diverse skill levels.
- Developed a **ranking system** to assess student skills and track progress over time.
- Managed client interactions, including processing **payments, memberships**, and addressing **concerns** regarding curriculum.
- Coordinated meetings and communicated with parents and clients to ensure **smooth operations and satisfaction**.

EDUCATION

Simon Fraser University | Burnaby, BC

Sept 2024 - May 2027

Bachelor's of Science in Computer Science

Langara College | Vancouver, BC

Sept 2022 - Apr 2024

Completed coursework in Computer Science and Mathematics

ACCOMPLISHMENTS

- **Bronze Medalist in Taekwondo at World University Games (Germany, 2025)**
 - **First-ever Canadian medalist** in Taekwondo at the World University Games
 - One of **three male athletes** to represent **Canada**
 - Competed against athletes from **14 countries**